第1章 Orange 簡介

大數據分析

- R/Python/Julia/SQL程式設計與應用 (R/Python/Julia/SQL Programming and Application)
- 資料視覺化 (Data Visualization)
- 機器學習 (Machine Learning)
- 統計品管 (Statistical Quality Control)
- 最佳化 (Optimization)

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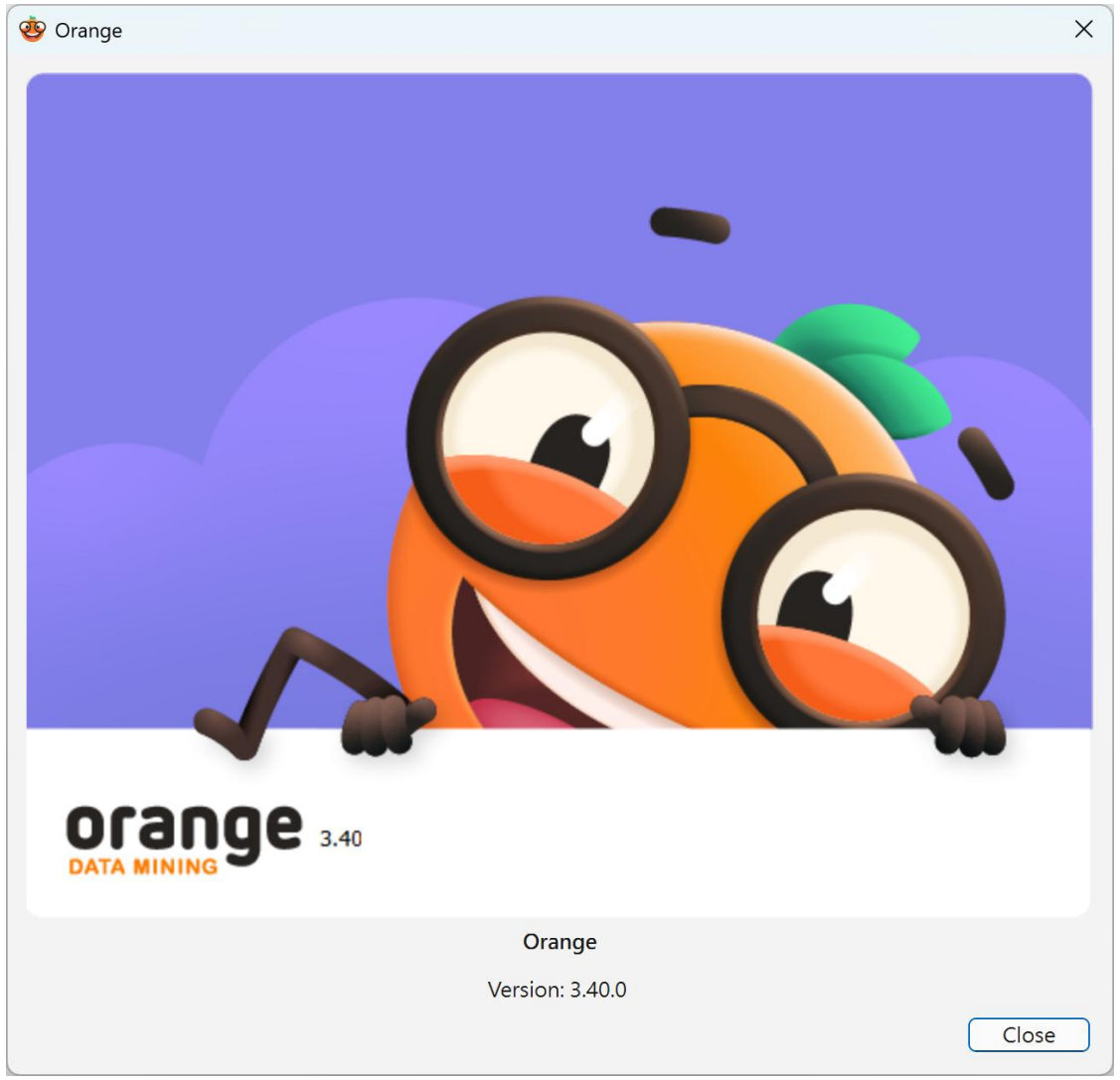
大綱

1. Orange 簡介
2. Orange 安裝
3. Orange 使用
4. Q & A



1. Orange 簡介

Orange 特性

- University of Ljubljana, Slovenia, 10 October 1996
- 視覺化程式設計工具(Visual Programming Tools)-**不用寫程式**
- 執行更加快速(C++)與視覺化操作
- 提供多種機器學習模組
- 開放原始碼與跨平台
- 適用於Python模組
- 資料預處理
- 模型訓練
- 部署

- 官網 https://orangedatamining.com/
- 元件 https://orangedatamining.com/widget-catalog/
- Tutorial https://orange3.readthedocs.io/projects/orange-data-mining-library/en/latest/

Orange3 add-ons (外掛元件)

1. **Orange3-Associate** **關聯規則**
2. Orange3-Bioinformatics 生物資訊
3. Orange3-Educational 機器學習教育示範
4. Orange3-Explain 重要特徵解釋
5. Orange3-Fairness 公平感知 AI 工具
6. Orange3-Geo 地理資料視覺化
7. Orange3-ImageAnalytics 影像分析
8. Orange3-Network 網路分析
9. Orange3-Prototypes 原型
10. Orange3-Pumice 教育活動
11. Orange3-SingleCell 單細胞分析
12. Orange-Spectroscopy 光譜分析
13. Orange3-Survival-Analysis 存活分析
14. Orange3-Text 文字探勘
15. Orange3-Textable 文字探勘, 編碼
16. Orange3-Timeseries 時間序列
17. Orange3-WorldHappiness 世界社會經濟指標
18. Orange-SNOM 掃描近場光學顯微術

https://github.com/biolab/orange3-addons/blob/master/OFFICIAL_ADDONS.txt

2. Orange 安裝

步驟1 下載 windows 免安裝檔

- https://orangedatamining.com/download/#windows
- Orange3-3.40.0.zip (476MB)

Windows

Standalone installer (default)

Orange3-3.40.0-x86_64.exe

Can be used without administrative privileges.

Portable Orange

Orange3-3.40.0.zip

No installation needed. Just extract the archive and open the shortcut in the extracted folder.

採用免安裝版本

下載 macOS, Linux 版本

- https://orangedatamining.com/download/

macOS

Orange for Apple silicon

Orange3-3.40.0-Python3.12.10-arm64.dmg

Orange for Intel

Orange3-3.40.0-Python3.12.10-x86_64.dmg

Not sure which installer to select? Click the Apple logo in the top-left corner of your screen, select About This Mac, and check the Chip or Processor field. If you see Apple, select the Orange for Apple Silicon installer. If you see Intel, select the Orange for Intel.

Other platforms

Anaconda

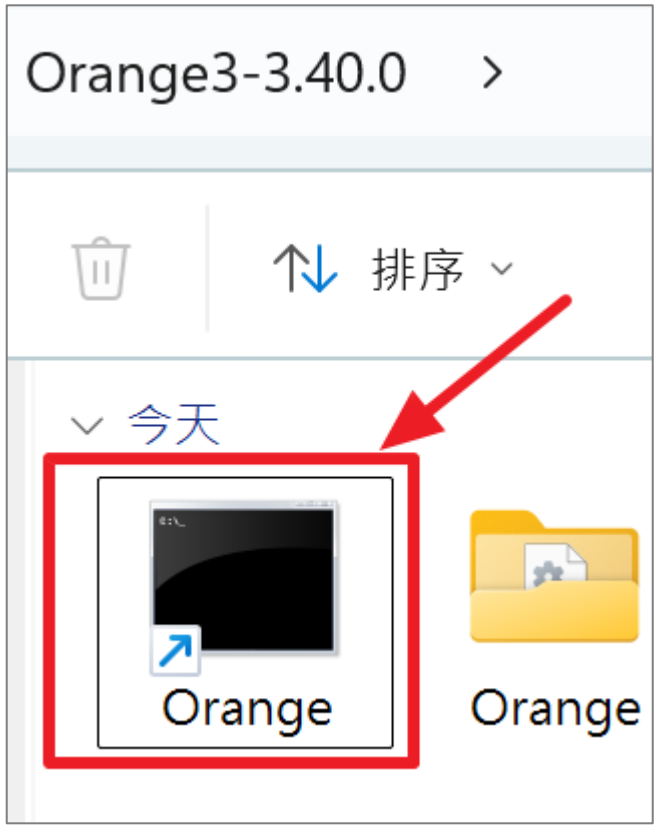
Create and activate a conda environment for Orange (optional, but recommended)

```
conda create python=3.11 --yes --name orange3
conda activate orange3
```

步驟2 解壓縮 zip 檔案

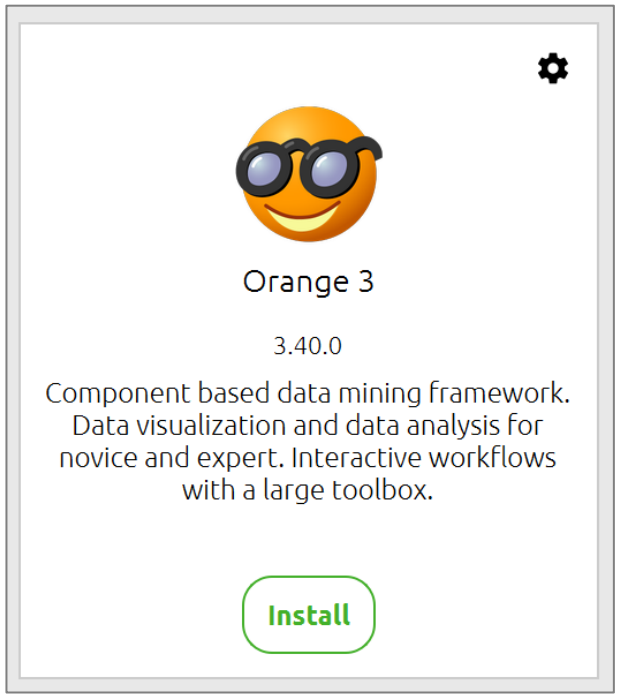
開啟 Orange

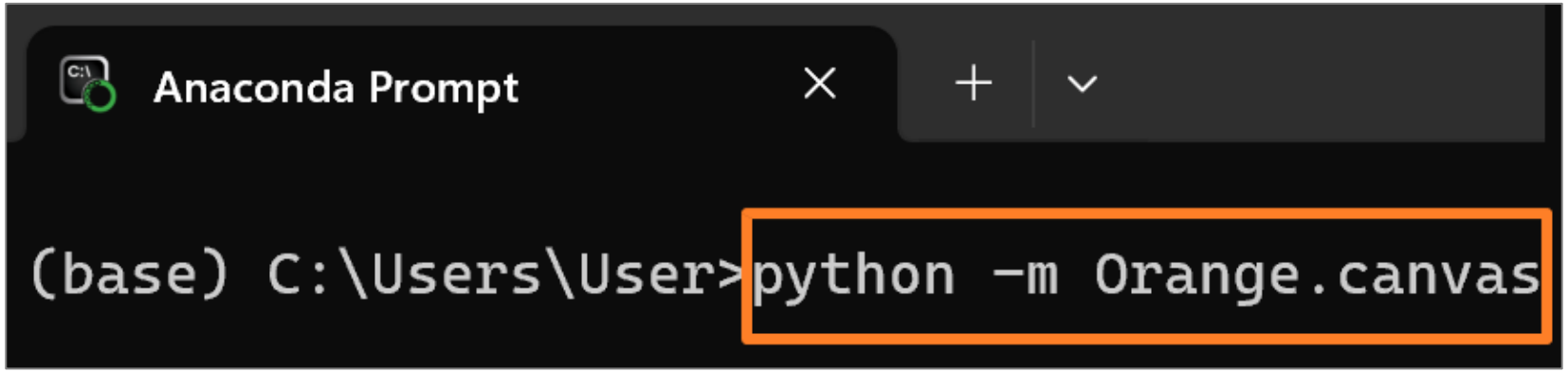
- 開啟方法1. 開啟 Orange 檔案 --> 教學採用此方法



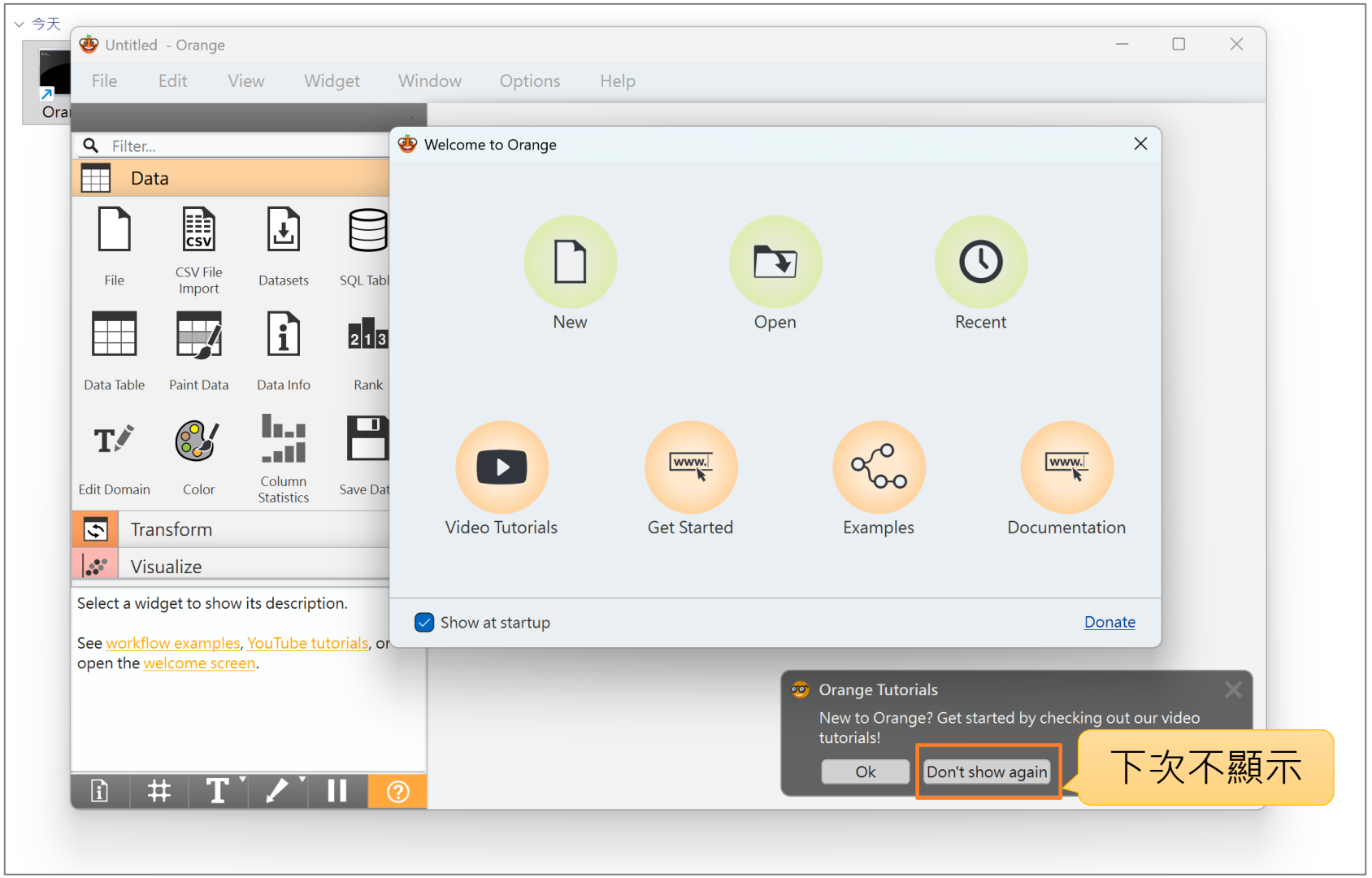
開啟 Orange

- 開啟方法2: 考慮已經安裝完成二大元件:
  - Anaconda
  - 使用 Anaconda Navigator 安裝完成 Orange
- 命令提示字元輸入: python -m Orange.canvas





Orange 初始畫面


今天
Untitled - Orange
File
Edit
View
Widget
Window
Options
Help
Filter...
Data
File
CSV File Import
Datasets
SQL Tabl
Data Table
Paint Data
Data Info
Rank
Edit Domain
Color
Column Statistics
Save Dat
Transform
Visualize
Select a widget to show its description.
See workflow examples, YouTube tutorials, or open the welcome screen.
Welcome to Orange
New
Open
Recent
Video Tutorials
Get Started
Examples
Documentation
Show at startup
Donate
Orange Tutorials
New to Orange? Get started by checking out our video tutorials!
Ok
Don't show again
下次不顯示

收工囉

參考資料

- RWEPA: http://rwepa.blogspot.com/
- https://github.com/rwepa/teaching-Orange
- https://github.com/rwepa/orange3_associate
- 第 9 章 關聯規則應用, Python 程式設計-李明昌 免費電子書
  - http://rwepa.blogspot.com/2020/02/pythonprogramminglee.html

謝謝您的聆聽

Q & A

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http://rwepa.blogspot.tw/